

Bits and digitization

Data is often in the news, whether it's data protection or data storage. But what exactly is data, and how do computers use it and store it?

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What is data?

Data is the name given to information that's processed or stored by a computer. It covers everything: from instructions for the computer to carry out, to text, music, photos, and more. No matter what it represents, digital data is stored as a series of binary numbers made up of the digits 1 and 0. Humans normally count in multiples of 10, in what is called the decimal system. The binary system, however, is based on multiples of two.

1 bit
0 or 1

▽ Counting bits and bytes

A single 1 or 0 is known as a binary information digit, or bit. A byte is 8 bits, and half a byte is a nibble. One byte can store a single character of text – for example, “!”, “a”, or “4”.

1 byte
One letter
of the
alphabet

1 kilobyte (KB)
One small
text email

1 megabyte (MB)
One good
quality
digital picture

1 gigabyte (GB)
240 pieces
of music

1 terabyte (TB)
100 hours
of video

1 petabyte (PB)
13.3 years of HDTV video

Why 0 and 1?

Modern computer chips contain billions of transistors that act like tiny switches. These transistors are used to represent binary numbers. If no electrical current is flowing through a transistor, it represents a 0 in binary, while flowing current represents a 1. Nowadays, chips include transistors that are 14 nanometres long – around 500 times smaller than a human blood cell.

▷ On or off

If the switch is turned off, no current flows through the circuit; if the switch is on, a current flows through and the bulb is lit.

An open switch stops the flow of current, and the bulb does not light up. This is a 0 in the binary system.

A closed switch allows the current to flow, causing the bulb to light up. This is a 1 in the binary system.

